

MULTI-STAGE GENAI PROMPT WORKFLOW

AI-Powered Study Planner for Programming Practice

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1. Scenario

The proposed system is an AI-assisted study planner for programming students. Its purpose is to convert practice history into a focused weekly plan, identify topics that need attention, recommend suitable exercises, and create short revision quizzes. The workflow uses three connected prompt stages: analysis, solution drafting, and review.

2. Sample Learner Profile

Topic	Practice Accuracy	Recent Attempts	Status
Programming Fundamentals	86%	25	Strong
Functions & Modules	72%	18	Developing
Lists / Dictionaries	61%	22	Needs practice
Exception Handling	48%	12	Priority
Object-Oriented Programming	55%	15	Priority

3. Stage 1 – Learner Analysis

Prompt objective: Ask the model to interpret the practice profile, rank topics by need, and propose a realistic study sequence. The output should distinguish strong skills from topics requiring intervention.

Illustrative result: Exception handling and OOP should receive the highest attention, while programming fundamentals can be maintained through short revision exercises. Lists and dictionaries should receive moderate practice focused on common mistakes.

4. Stage 2 – Plan Generation

Prompt objective: Convert the analysis into a seven-day learning schedule. Each day should contain a concept target, a small coding exercise, a short quiz, and a measurable completion condition. The schedule should avoid assigning advanced topics before prerequisite concepts are reviewed.

Proposed output structure:

Day	Focus	Activity	Check
1	Exceptions	try/except practice	5-question quiz
2	Exceptions	custom exception exercise	error-handling checklist
3	OOP basics	class and object exercise	concept quiz
4	OOP methods	inheritance practice	coding task
5	Collections	dictionary/list problems	timed practice
6	Mixed revision	two mini programs	self-test
7	Assessment	combined challenge	score + review

5. Stage 3 – Critical Review

The final prompt asks a reviewer model to inspect the generated plan for four risks: unsuitable difficulty, missing prerequisites, excessive daily workload, and vague success criteria. The reviewer should identify problems first

and then return a corrected schedule. This makes the third stage a quality-control gate rather than another blind generation step.

6. Refined Workflow

Practice History → Performance Analysis → Priority Topics → Weekly Plan → Exercise Generation → Quiz → Performance Update → Next-Week Adjustment

7. Expected GenAI Features

Feature	Purpose
Performance interpretation	Turn raw practice results into understandable learning priorities.
Adaptive planning	Allocate more study time to low-performing topics.
Exercise generation	Create practice tasks at an appropriate difficulty.
Quiz creation	Produce short checks for concept retention.
Self-critique	Review the plan before presenting it as final.
Progress revision	Use new scores to adjust the next study cycle.

8. Conclusion

This workflow shows how chained prompting can be used to turn learner-performance data into an actionable study plan. The first stage interprets the learner profile, the second constructs a schedule, and the third checks the schedule for quality and feasibility. The result is a reusable process that can be updated whenever new practice results become available.